1.6.1 Some more special properties of square numbers

(i) The square of a natural number other than 1, is either a multiple of 3 or exceeds a multiple of 3 by 1.

(ii) The square of a natural number, other than 1, is either a multiple of 4 or exceeds a multiple of 4 by 1.

(iii) The remainder of a perfect square when divided by 3, is either 0 or 1 but never 2.

(iv) The remainder of a perfect square, when divided by 4, is either 0 or 1 but never 2 and 3.

(v) When a perfect square number is divided by 8, the remainder is either 0 or 1 or 4, but will never be equal to 2, 3, 5, 6 or 7.



Perfect numbers such as 6, 28, 496, 8128 etc., are not square numbers.



Note

If a perfect square number ends in zero, then it must end with even number of zeroes always. We can verify this for a few numbers in the table given below.

Number	10	20	30	40	...	90	100	110	...	2000	...
Its Square	100	400	900	1600	...	8100	10000	12100	...	4000000	...



Think

Consider the claim: "Between the squares of the consecutive numbers n and $(n+1)$, there are $2n$ non-square numbers". Can it be true? How many non-square numbers are there between 2500 and 2601? Verify the claim.

1.7 Square Root

Squaring a number is another mathematical operation just like addition, subtraction, multiplication etc., Most mathematical operations have 'inverse' (meaning 'opposite') operations. For example, subtraction is the inverse of addition, division is the inverse of multiplication etc., Squaring also has an inverse operation namely finding the Square root.

The square root of a number n, written $\sqrt{n}$ (or) $n^{\frac{1}{2}}$, is the number that gives n when multiplied by itself. For example, $\sqrt{81}$ is 9, because $9 \times 9 = 81$.

In the adjacent table, we have square roots of all the perfect squares starting from 1 to 100.

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